Lab -5

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Title: Filter Bank Design

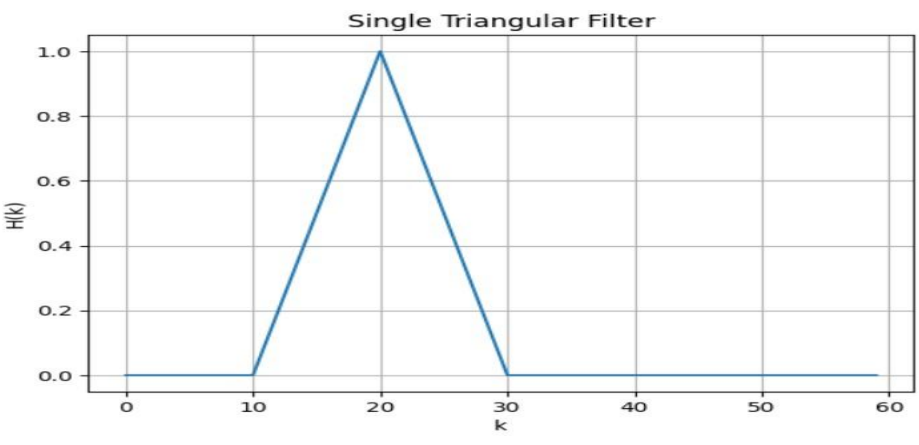
Objective :-

1. To understand how to construct a triangular filter
2. To learn how multiple triangular filters form a filter bank
3. To design and visualize a Mel Filter Bank
4. To understand the need for FFT bin mapping

Methodology :-

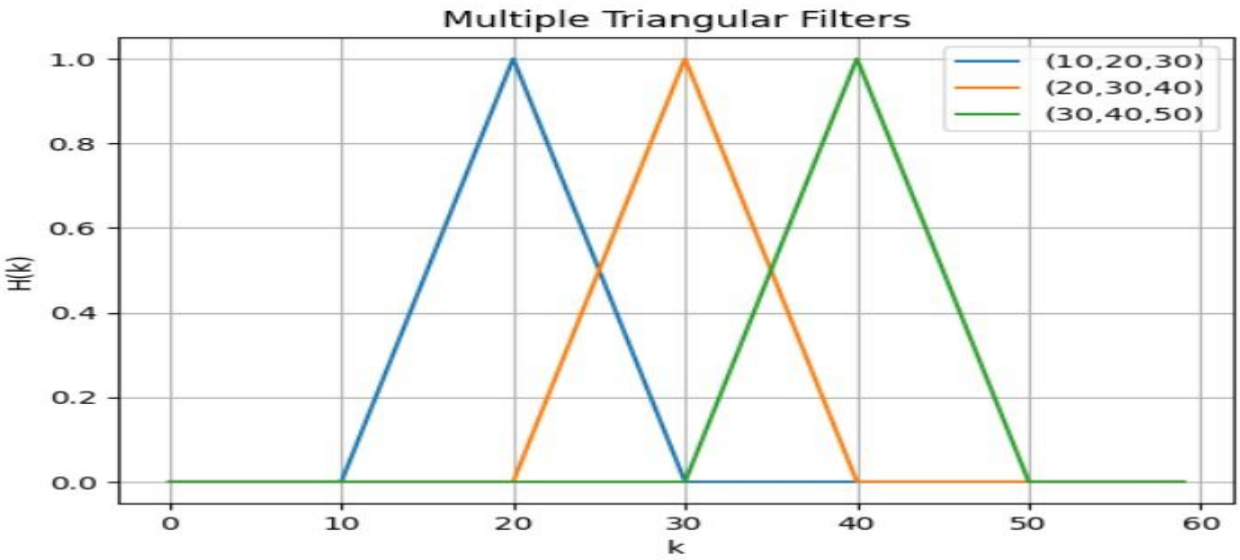
1. Defined parameters like sampling frequency (16 kHz), FFT size (512), and number of filters.
2. Constructed a single triangular filter using given points (a, b, c).
3. Created multiple triangular filters to form a filter bank.
4. Converted frequency values into Mel scale and generated Mel points.
5. Converted Mel points back to frequency and mapped them to FFT bins.
6. Built triangular filters using FFT bin values.
7. Plotted graphs for visualization (triangular filters, Mel points, filter bank).

RESULTS



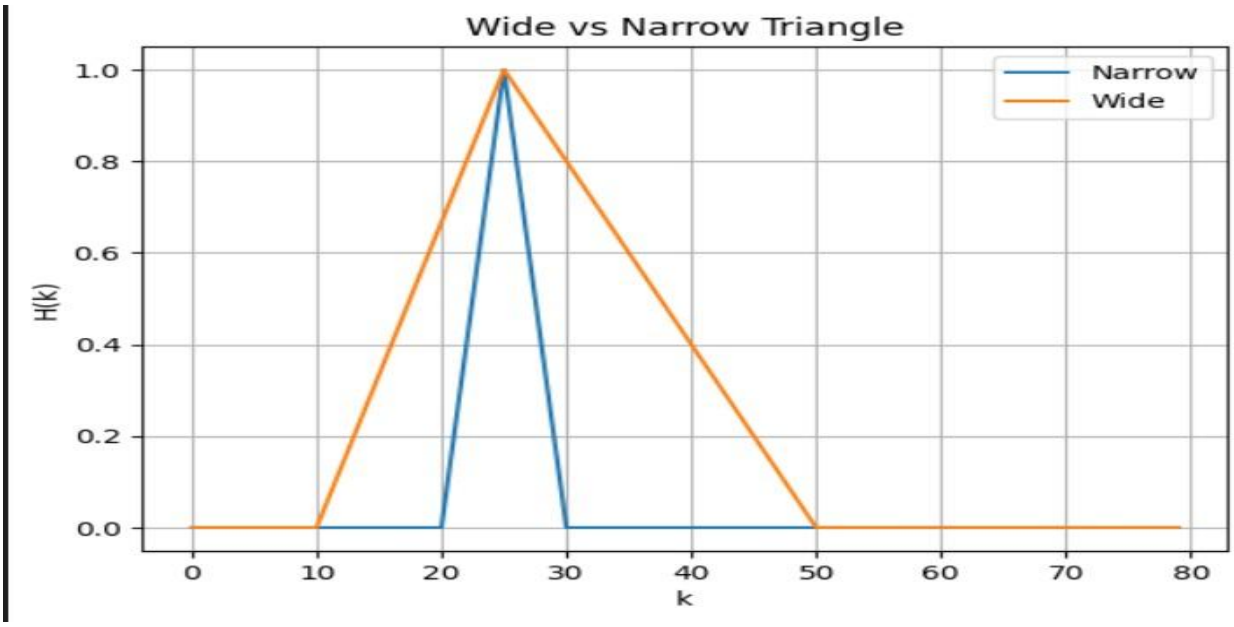
1. Single Triangular Filter Graph

The graph shows a triangular shape formed using three points (a, b, c).
The value increases linearly from point **a to b** and decreases from **b to c**. This represents how a filter passes frequencies around its center and suppresses others.



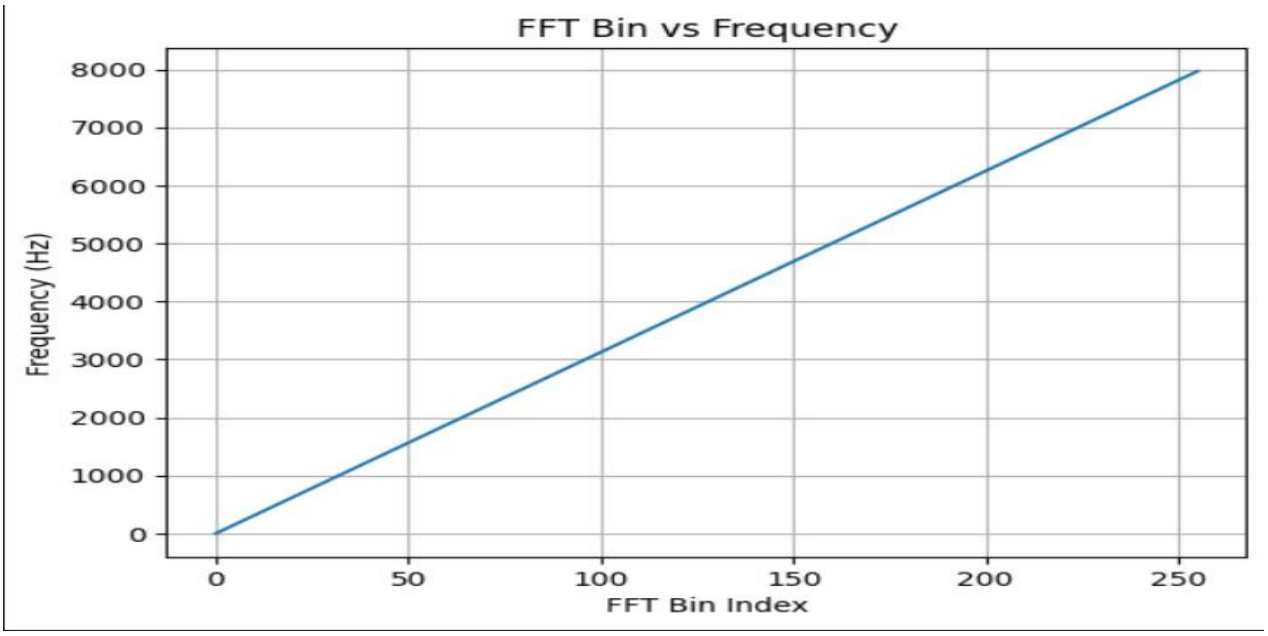
2. Multiple Triangular Filters

This graph shows several triangular filters placed next to each other.
Each filter overlaps slightly with neighboring filters to ensure smooth coverage of frequencies.
Together, they form a **filter bank**, which divides the signal into different frequency bands.



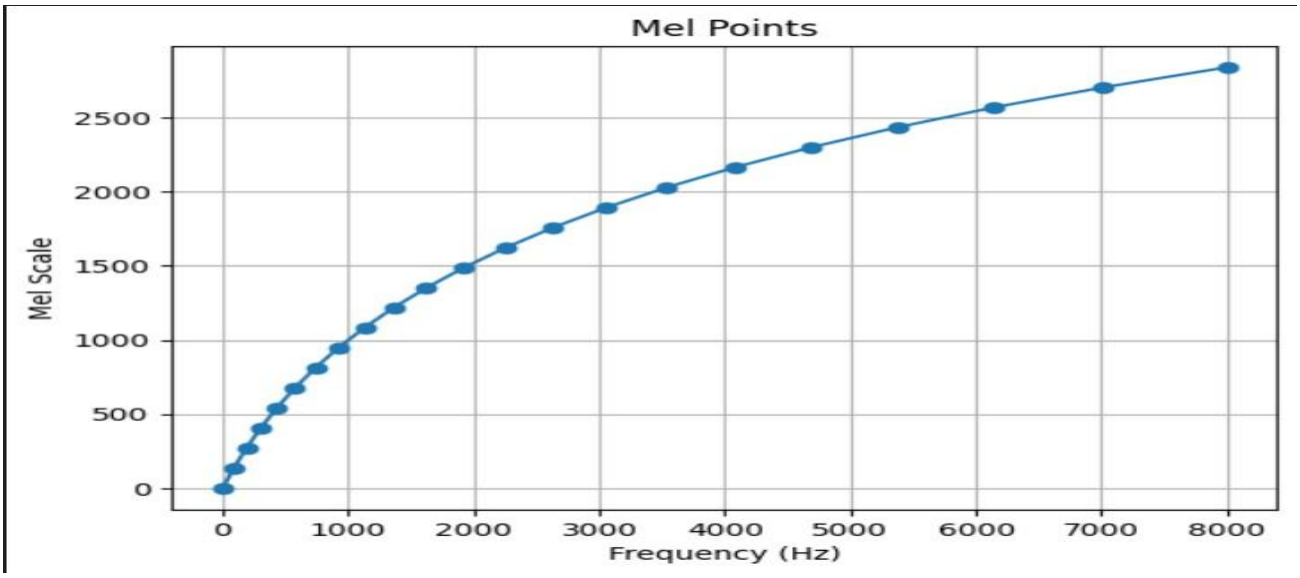
3. Wide vs Narrow Triangle

When the distance between (a, b, c) is increased, the triangle becomes wider. When the distance is reduced, the triangle becomes narrow and sharper. This shows how filter bandwidth affects frequency resolution.



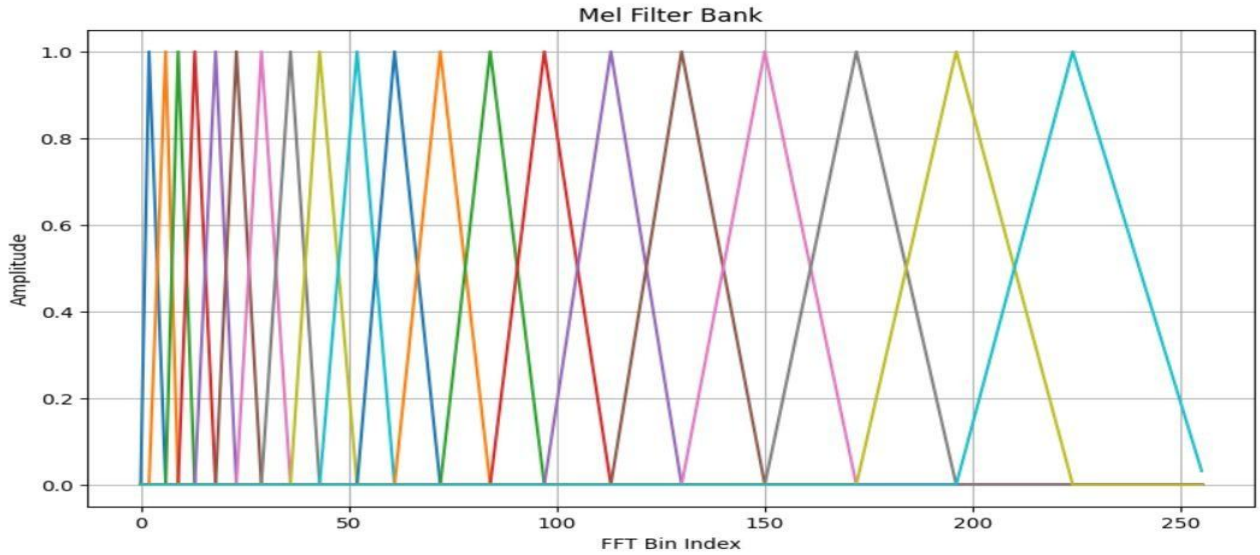
4. FFT Bin vs Frequency Plot

This graph shows the relationship between FFT bin index and actual frequency.
Each bin corresponds to a specific frequency value.
This mapping is important because filters are applied on FFT bins, not directly on frequency.



5. Mel Points Plot

Mel points are spaced according to human hearing perception.
Lower frequencies are closely spaced, while higher frequencies are more spread out. This helps in better representation of speech signals.



6. Mel Filter Bank Graph

This graph shows multiple triangular filters in Mel scale.
Each filter focuses on a specific frequency region.
Together, they form the Mel filter bank, which is widely used in speech processing (e.g., MFCC).

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Triangular Filter Points (a, b, c):
Filter 1: (0, 2, 6)
Filter 2: (2, 6, 9)
Filter 3: (6, 9, 13)
Filter 4: (9, 13, 18)
Filter 5: (13, 18, 23)
Filter 6: (18, 23, 29)
Filter 7: (23, 29, 36)
Filter 8: (29, 36, 43)
Filter 9: (36, 43, 52)
Filter 10: (43, 52, 61)
Filter 11: (52, 61, 72)
Filter 12: (61, 72, 84)
Filter 13: (72, 84, 97)
Filter 14: (84, 97, 113)
Filter 15: (97, 113, 130)
Filter 16: (113, 130, 150)
Filter 17: (130, 150, 172)
Filter 18: (150, 172, 196)
Filter 19: (172, 196, 224)
Filter 20: (196, 224, 256)
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7. Triangular Filter Points (a, b, c values)

The printed values represent the start, center, and end of each filter.
These points define how each triangular filter is constructed. They
are derived from Mel scale and mapped to FFT bins.

Effect of Changing Parameters

Changing number of filters, FFT size, or frequency range affects filter spacing.
More filters → better frequency resolution.
Fewer filters → less detail but simpler representation.

Conclusion

This experiment helped in understanding how triangular filters and Mel filter banks are constructed and used in speech processing. The results showed that Mel scale better represents human hearing by giving more importance to lower frequencies. The mapping of frequency to FFT bins ensures correct filter placement. Overall, this experiment provided a clear understanding of filter banks and their role in speech feature extraction.